# Individual Exercise 3 Report

## Objective:

The exercise focuses on building a writer identification system using the ICDAR17 dataset. It involves generating a codebook with k-means, encoding images with VLAD, and classifying them using SVMs.

## Dataset & Experimental Setup

- Dataset folders follow the ICDAR2017 Historical Writer Identification local-feature structure.
- Training descriptors: 1182 samples, Test descriptors: 3600 samples.
- Random seed fixed to 42 for reproducibility.
- Dictionary size K = 100 using MiniBatchKMeans.
- Approximately 500,000 training descriptors sampled to learn the codebook.
- Runs using --overwrite ensure a fair comparison between configurations.

**Results:**

| ***Experiment*** | ***Options*** | ***Top-1*** | ***mAP*** | ***Top-1 after Exemplar SVMs*** | ***mAP after Exemplar SVMs*** |
|---|---|---|---|---|---|
| Own SIFT (computed) | default VLAD | 0.82056 | 0.62995 | 0.88111 | 0.74187 |
| *Given descriptors* | default VLAD | 0.80583 | 0.61745 | 0.87556 | 0.73125 |
| *Baseline* | -clusters 32 + power norm | 0.83250 | 0.64483 | 0.89778 | 0.76466 |
| *GMP γ=1* | Gmp, gamma 1, power norm | 0.83861 | 0.65147 | 0.90389 | 0.77989 |
| *GMP γ=10* | Gmp, gamma 10,powernorm | 0.83472 | 0.64765 | 0.89833 | 0.77238 |
| *GMP γ=100* | Gmp, gamma 100, power norm | 0.83139 | 0.64336 | 0.89722 | 0.76524 |
| *Multi-VLAD + PCA* | Multivlad, codebooks 5, clusters 32, pcadim 1000 , power norm | | | 0.89000 | 0.74879 |

**Comparative Analysis:**

**Own descriptors vs given descriptors**

The goal was to compare the pipeline performance when using:

- SIFT descriptors computed by us from the images, and
- precomputed descriptors provided with the exercise.

results show that **own computed descriptors perform slightly better** than the given ones:

- Before Exemplar SVMs: mAP improves from **0.61745 → 0.62995**
- After Exemplar SVMs this improvement stays: **0.73125 → 0.74187**

This means our computeDescs + Hellinger normalization implementation is working correctly and produces competitive features.

**Part (f): Baseline VLAD vs GMP**

GMP improves over the baseline in almost all cases, especially after Exemplar SVMs:

- Baseline after Exemplar SVMs: **mAP = 0.76466**
- Best GMP (γ=1) after Exemplar SVMs: **mAP = 0.77989**

Among the gamma values i tried, **γ=1 is the best**. As gamma increases, performance slowly drops closer to baseline.

**Part (g): Multi-VLAD + PCA**

Multi-VLAD setup (5 codebooks, K=32, PCA=1000, powernorm) achieved:

- **Top-1 = 0.89**, **mAP = 0.74879** (final output line)

This is good, but it is **slightly below the best GMP run** after Exemplar SVMs (mAP 0.77989). A likely reason is PCA compression or that Multi-VLAD may need different settings (e.g., without PCA or different PCA dimension) to reach its best performance.

**Conclusion**

- my own extracted SIFT descriptors slightly outperform the provided descriptors, both before and after Exemplar SVMs.
- GMP improves results over baseline VLAD, and the best performance is obtained with **γ=1**.
- Multi-VLAD + PCA works and produces strong accuracy, but in current run it does not beat the best GMP configuration.

**Compare the effect of multi-VLAD with VLAD. Which feature benefits the most of it? What works better for you? E-SVM or multi-VLAD or a combination of both?**

From my comparison of results, standard VLAD with the right normalization already works strongly, and the biggest improvement i observed comes from using E-SVM rather than switching to multi-VLAD. In my runs, VLAD with power normalization followed by E-SVM reached about mAP 0.7647, while multi-VLAD with 5 codebooks (K=32) plus PCA to 1000 dimensions and power normalization followed by E-SVM reached about mAP 0.7488, meaning multi-VLAD did not outperform VLAD in current setup. Multi-VLAD is designed to benefit the VLAD representation itself by reducing quantization errors and capturing more diverse visual patterns, but in our case that benefit seems reduced, likely because PCA compression and parameter choices removed useful information. Overall, what works best for me based on the numbers that I already measured is E-SVM combined with a stronger encoding choice like GMP (gamma 1) plus power normalization, which gave me highest performance (around mAP 0.7799), so E-SVM is the most helpful component, and the multi-VLAD + E-SVM combination works but is not better than VLAD + E-SVM for my experiments.

**Result Evidence:**

```
uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ source Venv_ex3/bin/activate
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Exercise3.py \
>   --in_train "icdar2017-training-binary" \
>   --in_test  "ScriptNet-HistoricalWI-2017-binarized" \
>   --suffix_train .png --suffix_test .jpg \
>   --n_clusters 32 \
>   --powernorm \
>   --overwrite
 load random descriptors for dictionary
100%|| 100/100 [00:54<00:00,  1.83it/s]
 loaded 499354 descriptors:
Generating codebook...
 compute dictionary
Codebook generated and saved.
 compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|| 3600/3600 [32:30<00:00,  1.85it/s]
VLAD encodings for training computed
 evaluate
Top-1 accuracy: 0.8325 - mAP: 0.6448330471980314
Computing VLAD encodings for training data...
100%|| 1182/1182 [10:44<00:00,  1.83it/s]
VLAD encodings for training computed.
 esvm computation
100%|| 3600/3600 [20:41<00:00,  2.90it/s]
 Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.8977777777777778 - mAP: 0.7646565817349964
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$
```

```
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Exercise3.py   --in_train "icdar2017-training-binary"   --in_test  "ScriptNet-HistoricalWI-2017-binarized"
   --suffix_train .png --suffix_test .jpg   --n_clusters 32   --powernorm   --overwrite
#train: 1182
 load random descriptors for dictionary
100%|| 100/100 [00:54<00:00,  1.85it/s]
 loaded 499354 descriptors:
Generating codebook...
 compute dictionary
Codebook generated and saved.
 compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|| 3600/3600 [33:24<00:00,  1.80it/s]
VLAD encodings for training computed
 evaluate
Top-1 accuracy: 0.8325 - mAP: 0.6448330471980314
Computing VLAD encodings for training data...
100%|| 1182/1182 [10:46<00:00,  1.83it/s]
VLAD encodings for training computed.
 esvm computation
100%|| 3600/3600 [20:30<00:00,  2.93it/s]
 Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.8977777777777778 - mAP: 0.7646565817349964
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$
```

```
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Exercise3.py   --in_train "icdar2017-training-binary"   --in_test  "ScriptNet-HistoricalWI-2017-binarized"   --suffix_train
".png"   --suffix_test  ".jpg"   --overwrite
Codebook loaded from disk.
 compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for test data...
100%|| 3600/3600 [33:37<00:00,  1.78it/s]
VLAD encodings for test computed
 evaluate
Top-1 accuracy: 0.8205555555555556 - mAP: 0.6299521747049708
Computing VLAD encodings for training data...
100%|| 1182/1182 [11:06<00:00,  1.77it/s]
VLAD encodings for training computed.
 esvm computation
100%|| 3600/3600 [21:42<00:00,  2.76it/s]
 Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.8811111111111111 - mAP: 0.7418718618107301
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$
```

```
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Exercise3.py \
>   --in_train "/proj/ciptmp/sivichri/icdar17_local_features/train" \
>   --in_test  "/proj/ciptmp/sivichri/icdar17_local_features/test" \
>   --suffix_train "_SIFT_patch_pr.pkl.gz" \
>   --suffix_test  "_SIFT_patch_pr.pkl.gz" \
>   --overwrite
 load random descriptors for dictionary
100%|| 100/100 [00:05<00:00, 17.16it/s]
 loaded 484569 descriptors:
Generating codebook...
 compute dictionary
Codebook generated and saved.
 compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|| 3600/3600 [03:17<00:00, 18.21it/s]
VLAD encodings for training computed
 evaluate
Top-1 accuracy: 0.8058333333333333 - mAP: 0.617453263489768
Computing VLAD encodings for training data...
100%|| 1182/1182 [01:00<00:00, 19.54it/s]
VLAD encodings for training computed.
 esvm computation
100%|| 3600/3600 [12:23<00:00,  4.84it/s]
 Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.8755555555555555 - mAP: 0.7312484603025778
```

```
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Exercise3.py   --in_train "icdar2017-training-binary"   --in_test  "ScriptNet-HistoricalWI-2017-binarized"
   --suffix_train .png --suffix_test .jpg   --multivlad --n_codebooks 5 --n_clusters 32   --pca_dim 1000   --powernorm   --overwrite
[codebook 1/5] sampling descriptors...
[codebook 1] clustering...
[codebook 2/5] sampling descriptors...
100%|| 100/100 [00:53<00:00,  1.88it/s]
[codebook 2] clustering...
[codebook 3/5] sampling descriptors...
100%|| 100/100 [00:54<00:00,  1.84it/s]
[codebook 3] clustering...
[codebook 4/5] sampling descriptors...
100%|| 100/100 [00:53<00:00,  1.86it/s]
[codebook 4] clustering...
[codebook 5/5] sampling descriptors...
100%|| 100/100 [00:54<00:00,  1.84it/s]
[codebook 5] clustering...
  VLAD for codebook 1/5
100%|| 3600/3600 [31:18<00:00,  1.92it/s]
  VLAD for codebook 2/5
100%|| 3600/3600 [32:07<00:00,  1.87it/s]
```

```
  warnings.warn(
100%|| 3596/3600 [20:44<00:01,  2.87it/s]
/usr/lib/python3/dist-packages/sklearn/svm/_classes.py:31: FutureWarning: The default value of `dual` will change from `True` to `'auto'` in 1.5. Set the value
of `dual` explicitly to suppress the warning.
  warnings.warn(
100%|| 3597/3600 [20:44<00:01,  2.75it/s]
/usr/lib/python3/dist-packages/sklearn/svm/_classes.py:31: FutureWarning: The default value of `dual` will change from `True` to `'auto'` in 1.5. Set the value
of `dual` explicitly to suppress the warning.
  warnings.warn(
100%|| 3598/3600 [20:45<00:00,  2.85it/s]
/usr/lib/python3/dist-packages/sklearn/svm/_classes.py:31: FutureWarning: The default value of `dual` will change from `True` to `'auto'` in 1.5. Set the value
of `dual` explicitly to suppress the warning.
  warnings.warn(
100%|| 3599/3600 [20:45<00:00,  2.92it/s]
/usr/lib/python3/dist-packages/sklearn/svm/_classes.py:31: FutureWarning: The default value of `dual` will change from `True` to `'auto'` in 1.5. Set the value
of `dual` explicitly to suppress the warning.
  warnings.warn(
100%|| 3600/3600 [20:45<00:00,  2.89it/s]
 Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.9038888888888889 - mAP: 0.7798866739464215
```

```
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Exercise3.py --in_train icdar2017-training-binary --in_test ScriptNet-HistoricalWI-2017-binarized \
>   --suffix_train .png --suffix_test .jpg --gmp --gamma 1 --powernorm --overwrite
Codebook loaded from disk.
 compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|                                                                                                                                                  | 3600/3600 [38:01<00:00,  1.58it/s]
VLAD encodings for training computed
 evaluate
Top-1 accuracy: 0.8386111111111111 - mAP: 0.6514747313775034
Computing VLAD encodings for training data...
100%|                                                                                                                                                  | 1182/1182 [12:06<00:00,  1.63it/s]
VLAD encodings for training computed.
 esvm computation
100%|                                                                                                                                                  | 3600/3600 [20:37<00:00,  2.91it/s]
 Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.9038888888888889 - mAP: 0.7798873938053139
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$
```

```
uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Exercise3.py   --in_train icdar2017-training-binary   --in_test ScriptNet-HistoricalWI-2017-binarized   --suffix_train .png --suffix_te
st .jpg   --gmp --gamma 10 --powernorm --overwrite
Codebook loaded from disk.
 compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|                                                                                                                                                  | 3600/3600 [31:35<00:00,  1.90it/s]
VLAD encodings for training computed
 evaluate
Top-1 accuracy: 0.8347222222222223 - mAP: 0.6476537271152145
Computing VLAD encodings for training data...
100%|                                                                                                                                                  | 1182/1182 [10:08<00:00,  1.94it/s]
VLAD encodings for training computed.
 esvm computation
  0%|                                                                                                                                                  | 0/3600 [00:00<?, ?it/s]
```

```
  warnings.warn(
100%|                                                                                                                                                  | 3600/3600 [21:16<00:00,  2.82it/s] Evaluate new enco
dings after Exemplar SVMs
Top-1 accuracy: 0.8983333333333333 - mAP: 0.7723753482151818
```

```
(Venv_ex3) uw64eqyr@cip4a0:~/Desktop/CV/Ex-3$ python3 Exercise3.py --in_train icdar2017-training-binary --in_test ScriptNet-HistoricalWI-2017-binarized   --suffix_train .png --s
ffix_test .jpg --gmp --gamma 100 --powernorm --overwrite
Codebook loaded from disk.
 compute VLAD for test
VLAD encodings computed.
#test: 3600
Computing VLAD encodings for training data...
100%|                                                                                                                                                  | 3600/3600 [37:20<00:00,  1.61it/s
VLAD encodings for training computed
 evaluate
Top-1 accuracy: 0.8313888888888888 - mAP: 0.6433631227431028
Computing VLAD encodings for training data...
100%|                                                                                                                                                  | 1182/1182 [11:54<00:00,  1.66it/s
VLAD encodings for training computed.
 esvm computation
100%|                                                                                                                                                  | 3600/3600 [21:19<00:00,  2.81it/s
 Evaluate new encodings after Exemplar SVMs
Top-1 accuracy: 0.8972222222222223 - mAP: 0.7652435029785227
```

```
100%|                                                                                                                                                  | 3600/3600 [01:21<00:00, 44.23it/s]
Top-1 accuracy: 0.89 - mAP: 0.7487926714312304
```